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Dataset: [Red Wine Quality](#)

I. Goals and key insights

For this dataset about red wine quality, we aim to answer these two questions: “What features affect red wine quality?” and “What methods should be used to predict quality?”

1. What features affect red wine quality?

The analysis of boxplots, the correlation matrix, and a linear regression model reveals several key attributes that influence red wine quality. The boxplot analysis indicates that lower volatile acidity, higher levels of sulphates, and increased alcohol content are associated with higher wine quality ratings. In contrast, fixed acidity, residual sugar, and pH do not show a consistent correlation, suggesting they may be less critical in determining quality. The correlation matrix further supports these findings, highlighting strong positive relationships between alcohol and sulphates with wine quality, while volatile acidity exhibits a strong negative correlation. Citric acid also shows a positive association, indicating its role in enhancing flavor and overall quality. Additionally, the linear regression model identifies volatile acidity, chlorides, and density as key negative influencers, while sulphates and alcohol are recognized as significant positive contributors.

However, it is not yet clear how the attributes specifically impact quality. Additionally, there are many other methods that have not been explored or applied in this analysis. Although we have identified influential attributes, we have not yet selectively used them for prediction or classification purposes but use all the attributes instead.

2. What methods should be used to predict quality?

To predict the numerical quality, we used a linear regression model, which suggests that there are some relationships between the predictor variables and wine quality. However, the model is not a perfect fit, as evidenced by the relatively high values of the error metrics.

To classify the quality as either good or bad, we employed three models: Logistic Regression, Random Forest, and K-Nearest Neighbors. While all models show high accuracy (around 87-88%), the accuracy metric alone is not sufficient to determine the best model for predicting good quality wine. The model with the highest accuracy, Logistic Regression and K-Nearest Neighbor (KNN) at 87.94%, may seem like the best choice. However, accuracy here can be misleading.

Despite high accuracy, the models struggle with predicting good quality wine. This is reflected in the low selectivity scores, especially for Logistic Regression and KNN, both of which have a selectivity of only 27.59%. This means that while the models are good at catching bad wine (high recall), they often fail to correctly identify good wine. Meanwhile, Random Forest, with the highest selectivity (34.48%), still shows that the correct identification of good wines is relatively low. This imbalance suggests that the models are biased toward predicting bad wine because bad cases are more frequent or easier to distinguish.

Therefore, in predicting whether wine is of good quality, the current models only work well when the wine quality is outstandingly bad or good. They don't reliably predict good wines under more nuanced conditions. The models perform better at identifying bad wines but often mislabel good wines as bad due to the lower selectivity rates

II. Data understanding

The dataset contains 1599 observations, including 11 attributes about wine characteristics and 1 output attribute (quality).

Explanation of features:

- fixed acidity: most acids involved with wine or fixed or nonvolatile (do not evaporate readily). Acidity is a characteristic determined by the total sum of acids that a sample contains. We can quantify the set of all of them in an undifferentiated way (total acidity) or in a grouped way (fixed acidity and volatile acidity). Fixed acidity corresponds to the set of low volatility organic acids such as malic, lactic, tartaric or citric acids and is inherent to the characteristics of the sample.
- volatile acidity: the amount of acetic acid in wine, which at too high of levels can lead to an unpleasant, vinegar taste. Volatile acidity corresponds to the set of short chain organic acids that can be extracted from the sample by means of a distillation process: formic acid, acetic acid, propionic acid and butyric acid.
- citric acid: found in small quantities, citric acid can add 'freshness' and flavor to wines. Citric acid is a colorless weak organic acid. It occurs naturally in citrus fruits. In biochemistry, it is an intermediate in the citric acid cycle, which occurs in the metabolism of all aerobic organisms.
- residual sugar: the amount of sugar remaining after fermentation stops, it's rare to find wines with less than 1 gram/liter. Residual sugar refers to the sugars left unfermented in a finished wine. It is measured by grams of sugar per liter (g/l). The amount of

residual sugar affects a wine's sweetness and, in the EU, the RS level is linked to specific labeling terms.

- chlorides: the amount of salt in the wine. The higher extraction of chloride during red winemaking is due to the ions extracted from skins during fermentation. Therefore, red juice should have no more than 356 mg/L chloride ions so that finished wine does not exceed the maximum legal level of 606 mg/L chloride ($356 \text{ mg/L in red juice} \times 1.7 = 606$).
- free sulfur dioxide: the free form of SO_2 exists in equilibrium between molecular SO_2 (as a dissolved gas) and bisulfite ion. What is free sulphur dioxide in wine? The free sulfites are those available to react and thus exhibit both germicidal and antioxidant properties. The bound sulfites are those that have reacted (both reversibly and irreversibly) with other molecules within the wine medium. The sum of the free and bound sulfites defines the total sulfite concentration.
- total sulfur dioxide: amount of free and bound forms of SO_2 ; in low concentrations, SO_2 is mostly undetectable in wine, but at free SO_2 . Simply put, Total Sulfur Dioxide (TSO_2) is the portion of SO_2 that is free in the wine plus the portion that is bound to other chemicals in the wine such as aldehydes, pigments, or sugars.
- density: the density of water is close to that of water depending on the percent alcohol and sugar content. How do you measure the density of wine? A hydrometer is an instrument used to measure liquid density. It is a sealed glass tube with a weighted bulb at one end, winemakers use this instrument to measure density of juice, fermenting wine and completed wine in relation to pure water. This ratio is called specific gravity (SG).
- pH: describes how acidic or basic a wine is on a scale from 0 (very acidic) to 14 (very basic); most wines are between 3-4. What is a high pH in wine? Wines which have higher pH values (>3.65) have a series of potential challenges during vinification and aging. First, high pH wines have an increased chance of microbial spoilage. Traditionally, sulfur dioxide (often in the form of potassium metabisulfite) is used to keep wines stable during aging.
- sulphates: a wine additive which can contribute to sulfur dioxide gas (SO_2) levels, which acts as an antimicrobial. Wine sulfites are naturally occurring at low levels in all wines, and are one of the thousands of chemical by-products created during the fermentation process. However, sulfites are also added by the winemaker to preserve and protect the wine from bacteria and yeast-laden invasions. For some, sulfur

allergies may be associated with headaches and stuffy sinuses after a glass or two of wine. It is a wine additive which can contribute to sulfur dioxide gas (SO₂) levels, which acts as an antimicrobial.

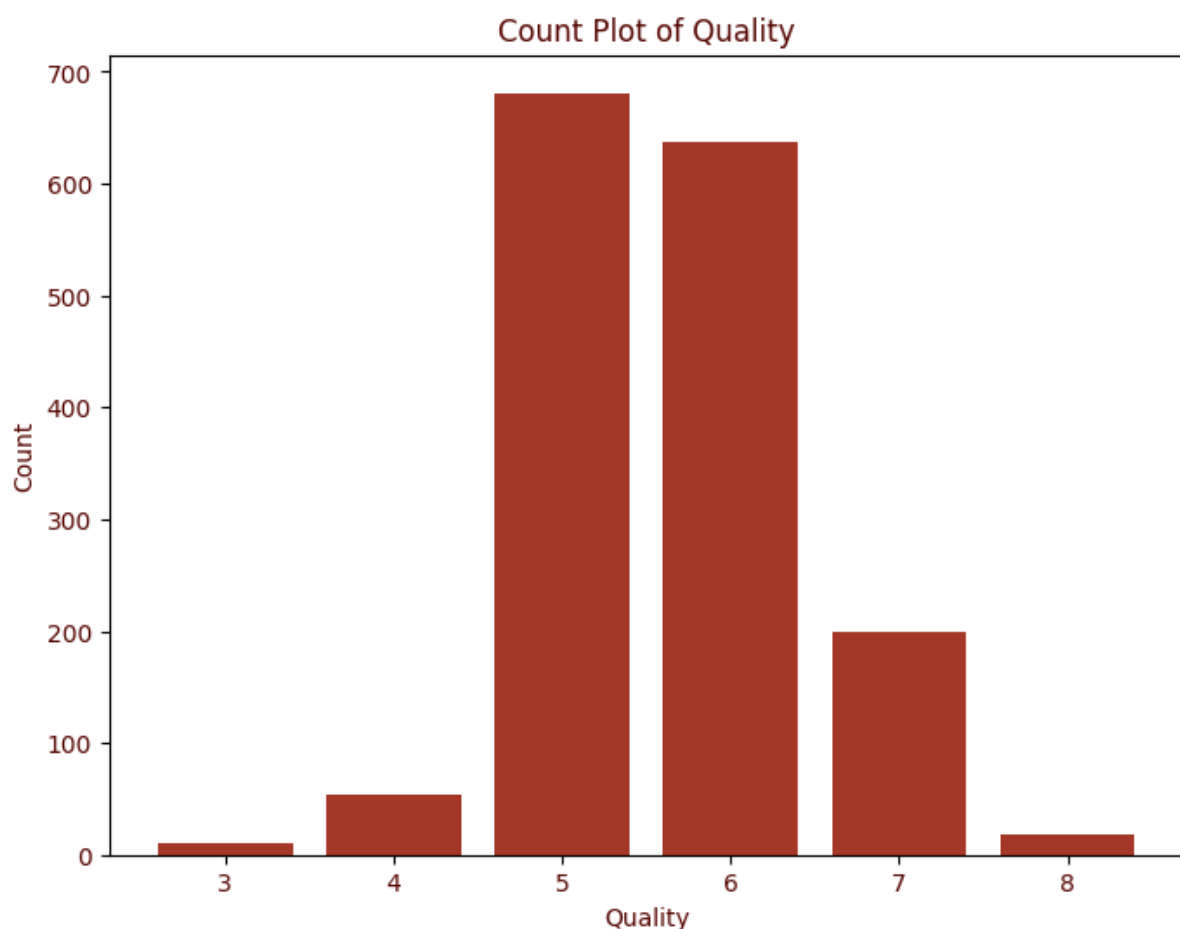
- Alcohol: this is the percent alcohol content of the wine
- quality: output variable (based on sensory data, score between 3 and 8).

III. Data preparation

There is no missing value, no duplicate in this dataset.

IV. EDA

1. Quality distribution

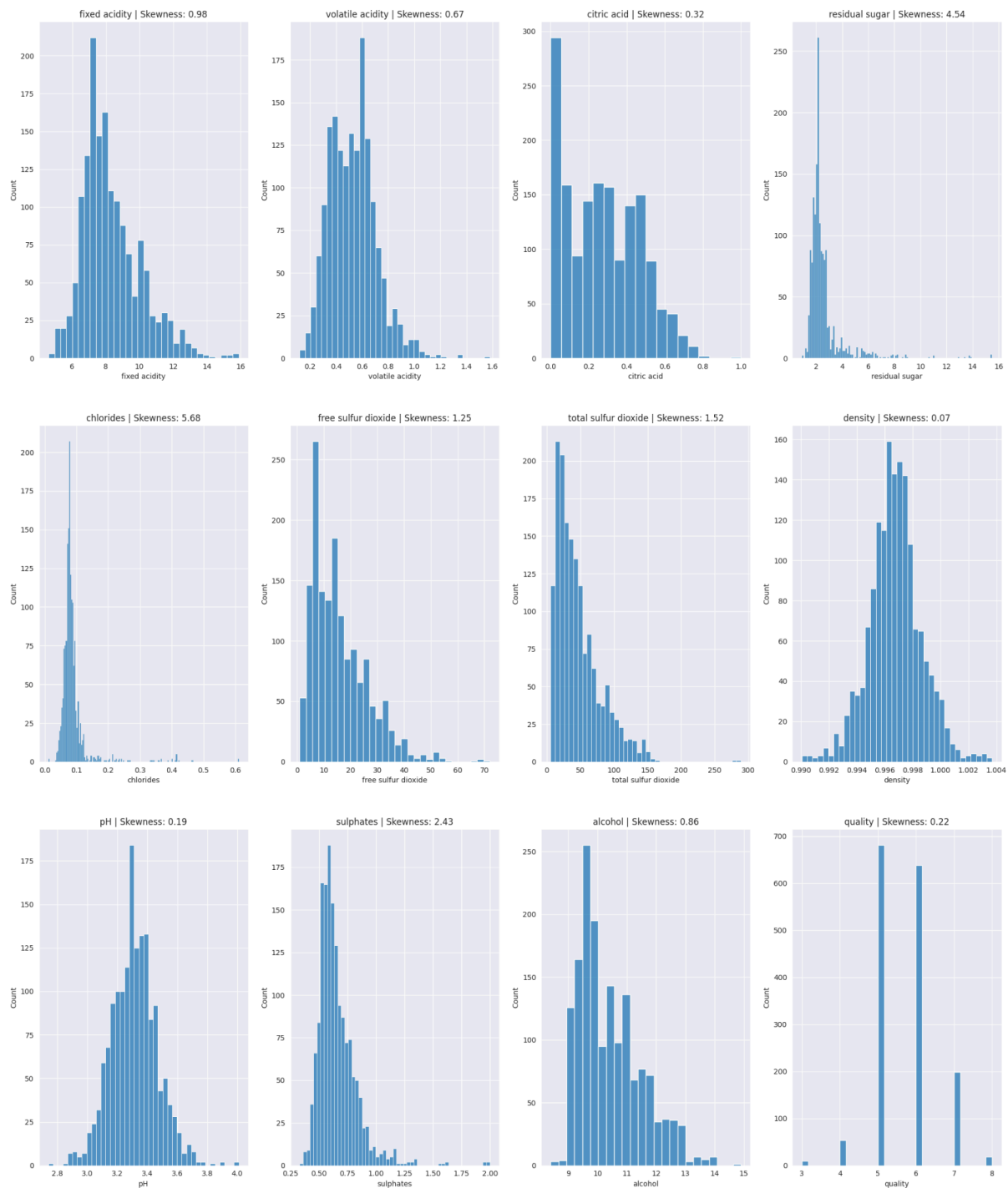


The histogram shows a clear distribution of wine quality (from 3 to 8). The distribution is skewed to the right, meaning there are more wines with higher quality ratings than lower quality ratings.

To create a binary classification problem, we established a threshold to separate "good" from "bad" quality wines. Given the distribution, a reasonable choice could be to consider wines

with a quality rating of 7 or 8 as "good" and those with ratings of 3-6 as "bad." This threshold aligns with the concentration of wines in the mid-range quality levels and provides a clear distinction between the two classes.

2. Other attributes distribution



The distribution of wine characteristics reveals several patterns that provide insights into the nature of the data. Many attributes, such as fixed acidity, volatile acidity, and citric acid,

display slight right-skewed distributions. For instance, fixed acidity values are clustered between 6 and 8, while volatile acidity is concentrated between 0.2 and 0.6, indicating that most wines have moderate levels of acidity. Citric acid is more normally distributed, with most values between 0.2 and 0.6, suggesting consistency in this characteristic across wines.

For example, the majority of wines have residual sugar values below 2, but a few outliers show significantly higher sugar content. Similarly, chlorides are mostly concentrated around low values, with only a small number of wines showing much higher chloride levels.

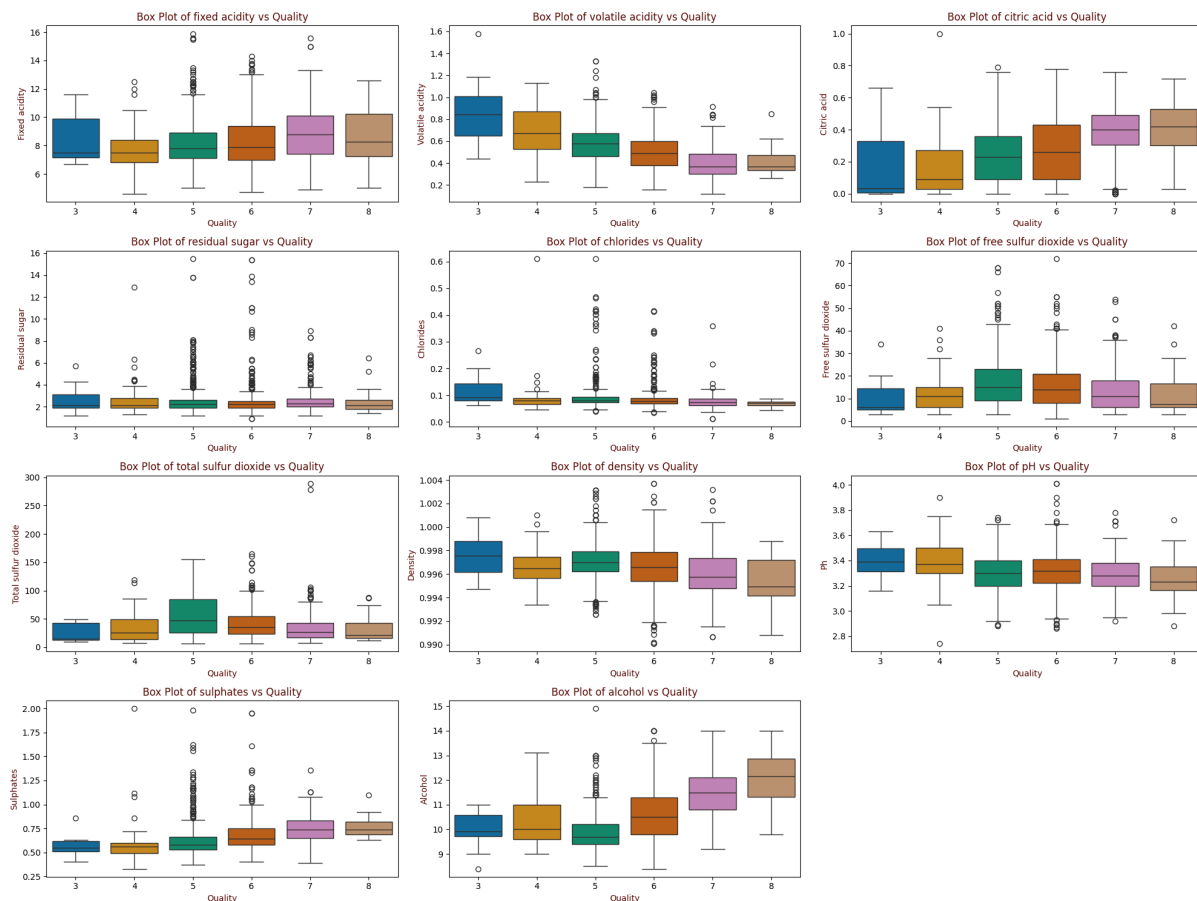
Other characteristics, such as free sulfur dioxide and total sulfur dioxide, also show right-skewed distributions, meaning most wines have lower levels of sulfur dioxide, while a few wines contain higher amounts. In contrast, density shows a nearly symmetrical distribution, indicating that most wines have similar density levels centered around typical values for wine.

Attributes like pH and alcohol are slightly skewed to the right, with pH values clustering around 3.0 to 3.4, and alcohol content ranging from 9% to 11% for most wines, with fewer wines having higher alcohol levels. Sulphates show a stronger right skew, with most wines having low sulphate levels and a few showing higher concentrations, which may influence quality perception.

Finally, wine quality itself is approximately normally distributed, with most wines rated as 5 or 6, and fewer wines receiving very low (3) or very high (8) ratings. This suggests that the majority of wines are of moderate quality, with outliers in both the lower and upper ends of the quality spectrum.

3. Boxplot

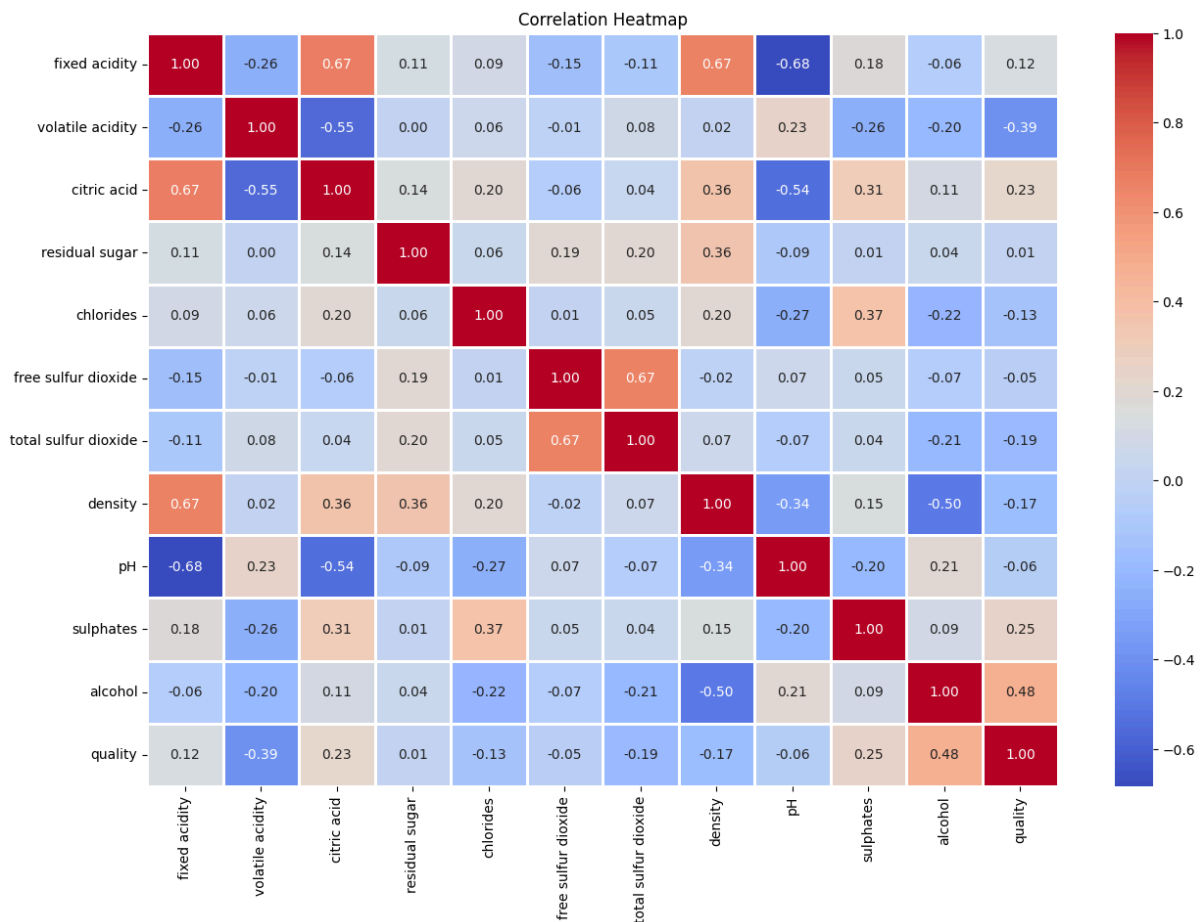
Box plots provide a visual summary of how the data for a variable is distributed. In this case, we are examining the distribution of various wine attributes across different quality levels.



The boxplot analysis reveals several key relationships between wine attributes and quality. First, volatile acidity shows a clear downward trend as wine quality improves, suggesting that lower volatile acidity is linked to higher-quality wines. Citric acid demonstrates a slight upward trend, indicating a potential positive association with wine quality, though the relationship is less pronounced. Sulphates show a more distinct increase, signaling a stronger positive correlation with quality. Alcohol, however, has the most significant increase, showing a strong positive relationship with wine quality.

Overall, the box plots suggest that red wines with lower volatile acidity, higher sulphate levels, and higher alcohol content tend to be rated higher in quality. Other characteristics, such as fixed acidity, residual sugar, and pH, do not display a clear or consistent correlation with quality, indicating that these factors may not play as pivotal a role in determining wine quality.

4. Correlation matrix



Several key attributes have a notable impact on wine quality. Alcohol shows a strong positive correlation, meaning that wines with higher alcohol content are generally rated as higher quality. Sulphates also exhibit a positive relationship, suggesting that increased sulphate levels contribute to better quality wine. In contrast, volatile acidity demonstrates a strong negative correlation, indicating that higher levels of volatile acidity are associated with lower quality. Citric acid, with its positive correlation, suggests it plays a significant role in enhancing wine flavor and overall quality.

Additionally, other important correlations exist between certain attributes. Fixed acidity and density have a strong positive relationship, implying that as fixed acidity increases, wine density also rises. There is a strong negative correlation between volatile acidity and citric acid, meaning that higher volatile acidity is often linked to lower citric acid levels. Lastly, density and alcohol content show a negative correlation, indicating that wines with higher density tend to have lower alcohol content. These correlations provide valuable insights into how various factors interplay to influence the overall quality of wine.

V. Model selection

We split the dataset into 1400 observations for training and 199 observations for testing.

1. Linear Regression

First, to do a regression task, we built a linear regression model.

```
Intercept (b): 12.67683706435222
Coefficients (slope): [ 2.11876808e-02 -1.05793606e+00 -1.48698571e-01  8.54070792e-03
-1.69475942e+00  5.18687422e-03 -3.49184007e-03 -8.82921987e+00
-3.26237146e-01  8.85243953e-01  2.76513540e-01]
MAE (Mean Absolute Error): 0.49608028255399983
MSE (Mean Squared Error): 0.3900419400228992
RMSE (Root Mean Squared Error): 0.6245333778293193
R^2 (R-squared): 0.41975766841296647
```

Intercept (b)	12.68	Free Sulfur Dioxide	0.0052
Fixed Acidity	0.0212	Total Sulfur Dioxide	-0.0035
Volatile Acidity	-1.0579	Density	-8.8292
Citric Acid	-0.1487	pH	-0.3262
Residual Sugar	0.0085	Sulphates	0.8852
Chlorides	-1.6948	Alcohol	0.2765

Key Negative Influencers:
Volatile acidity, Chlorides, and Density.

Key Positive Influencers:
Sulphates and alcohol.

While certain attributes like fixed acidity and residual sugar have minimal effects, their balance with the more impactful attributes is crucial for overall quality.

To evaluate this model, we calculate MAE, MSE, RMSE and R^2 .

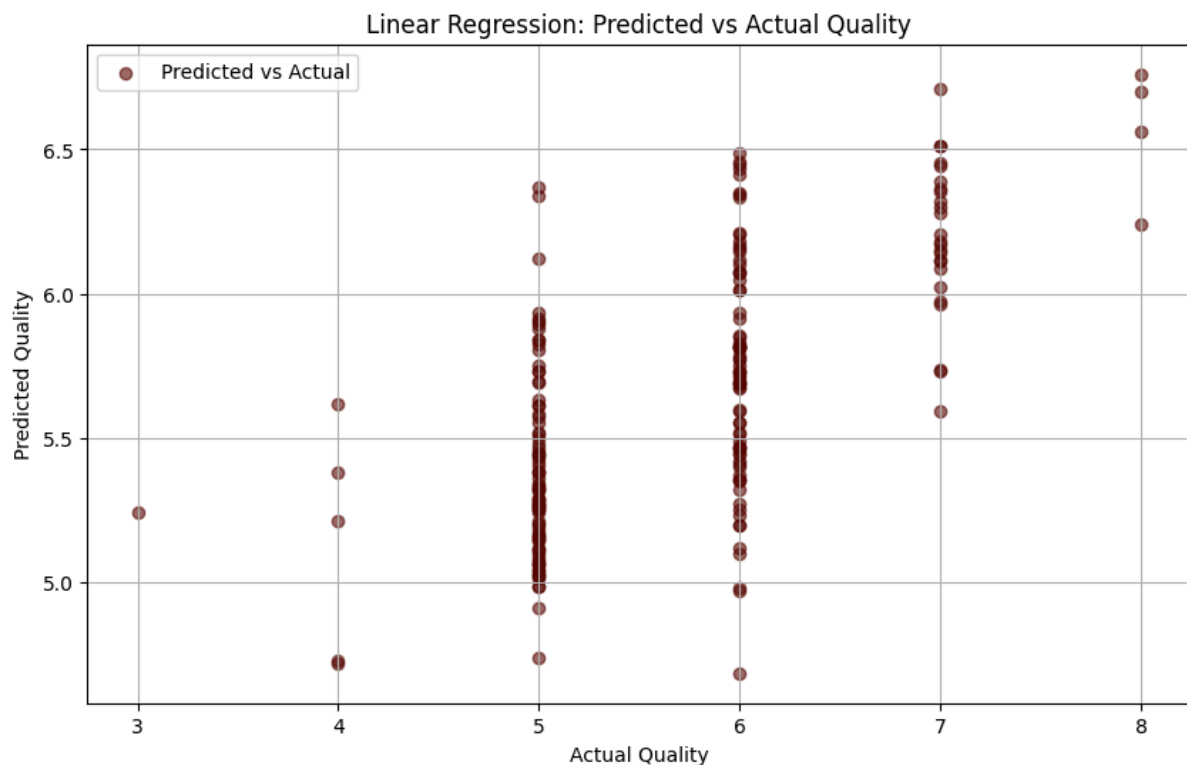
MAE measures the average absolute difference between the predicted and actual values. A lower MAE indicates better model performance. In this case, the MAE of 0.496 suggests that the model's predictions are generally within 0.496 units of the actual quality values.

MSE calculates the average squared difference between the predicted and actual values. It penalizes larger errors more heavily than MAE. An MSE of 0.390 indicates that the model's predictions are relatively close to the actual values, with some larger errors contributing to the overall error.

RMSE is the square root of the MSE. It provides an interpretable measure of the average error in the same units as the target variable (wine quality). An RMSE of 0.625 suggests that the model's predictions are, on average, 0.625 units away from the actual quality values.

R^2 measures the proportion of variance in the target variable (wine quality) that is explained by the model. An R^2 of 0.4198 indicates that the model explains approximately 42% of the variation in wine quality. This suggests that the model captures some of the underlying patterns in the data but is not a perfect fit.

Overall, the scatter plot and performance metrics suggest that the linear regression model is able to capture some of the relationship between the predictor variables and wine quality. However, the model is not a perfect fit, as evidenced by the relatively high values of the error metrics.



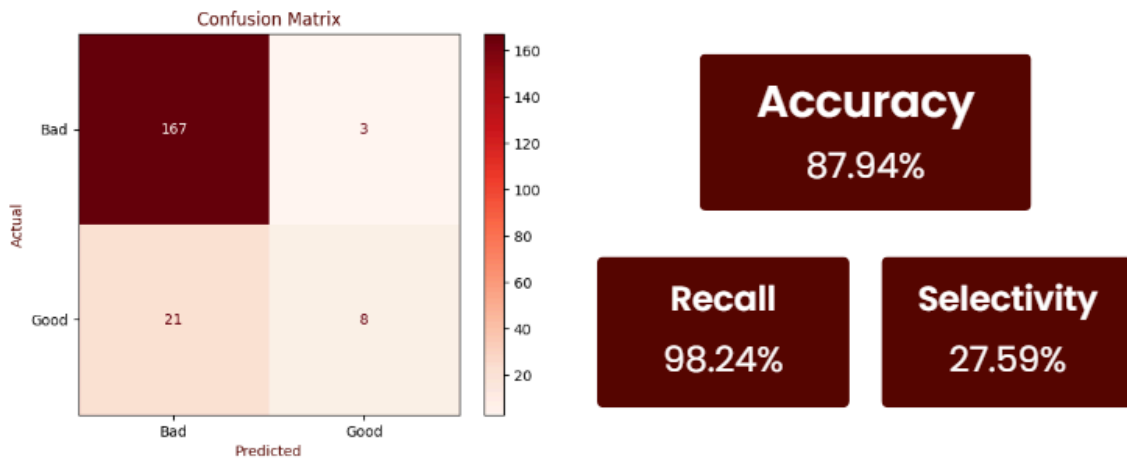
The scatter plot illustrates the relationship between the predicted wine quality values generated by a linear regression model and the corresponding actual quality ratings.

2. Logistic Regression

Second, we built the logistic regression model, which gives the equation to calculate the probability:

Logistic Regression Equation

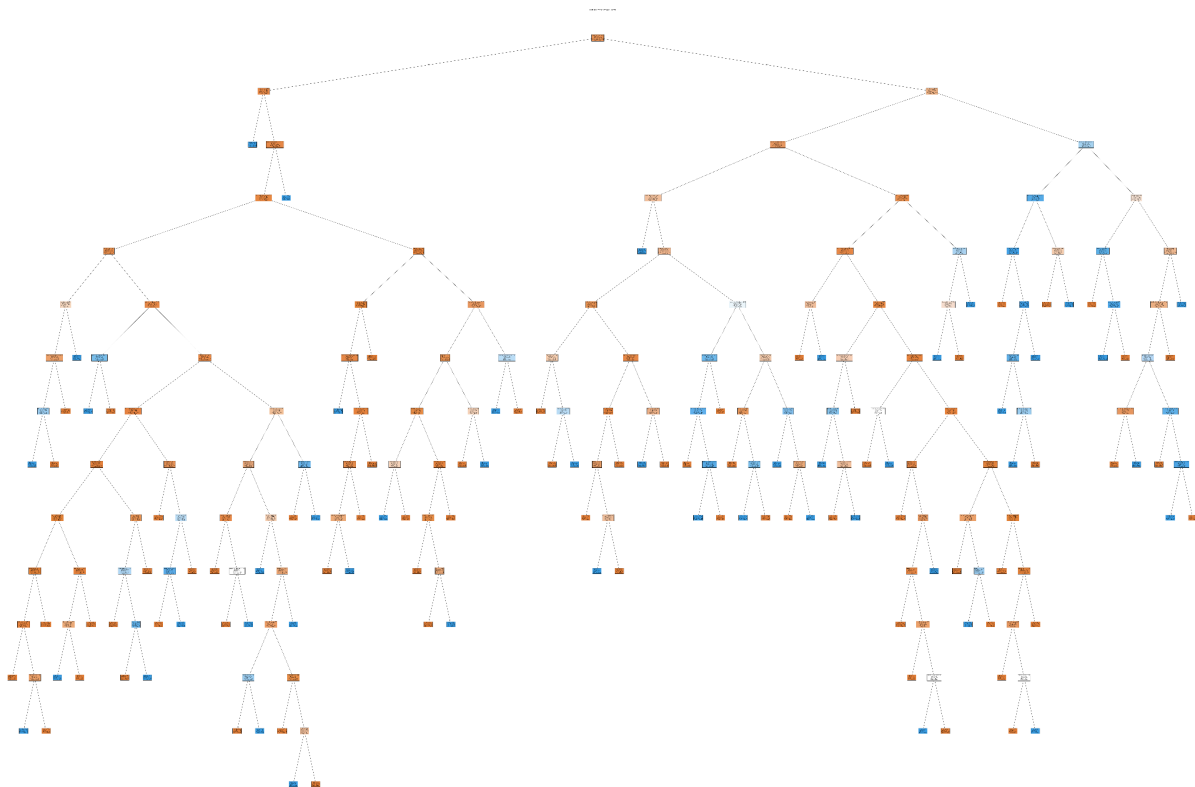
$$P = 1 / (1 + \exp(-(0.0783*x_0 + -2.3723*x_1 + 0.3924*x_2 + 0.0928*x_3 + -1.0236*x_4 + 0.0167*x_5 + -0.0185*x_6 + -0.0281*x_7 + -0.1339*x_8 + 2.4618*x_9 + 0.9619*x_{10} + -12.8775))))$$



The logistic regression model performs well in predicting "Bad" quality wines, with high recall. However, it struggles to accurately predict "Good" quality wines, with low selectivity. This suggests that the model may be biased towards predicting "Bad" quality wines.

3. Random Forest

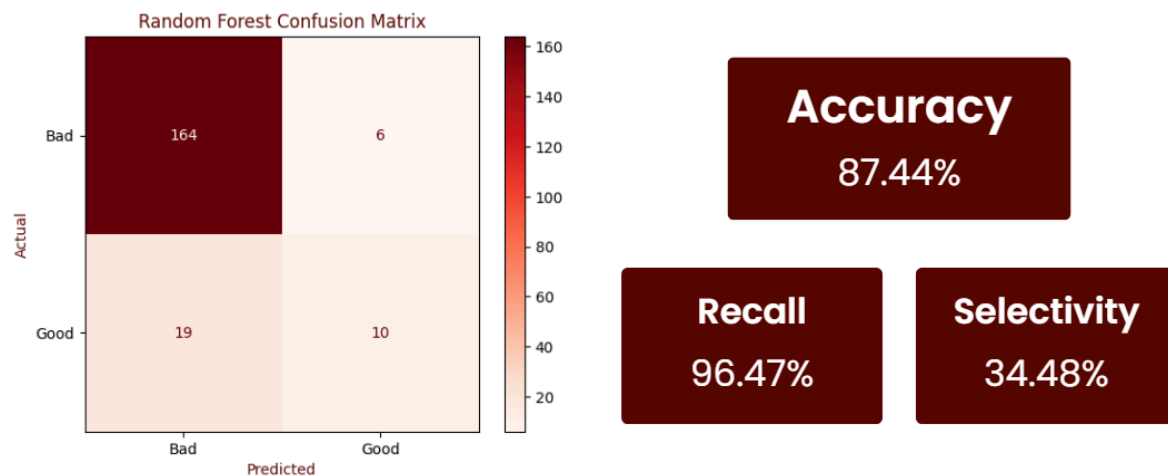
To do classification task, we used a random forest algorithm.



Example of the first tree in the Random Forest to visualize this dataset

In a Random Forest, each tree selects a random attribute (feature) at each split. For each attribute, it calculates the best split (e.g., Gini impurity or entropy) to classify data into "bad"

or "good". This process continues randomly for each tree, and the final prediction is based on the majority vote across all trees.



The model demonstrates an overall accuracy of 87.44%, which indicates that it is correctly predicting the quality of wines for the majority of cases. This model also shows strong recall, especially for the "Bad" wines, with a recall of 96.47%, indicating that it is highly effective at identifying wines that are actually "Bad". The lower selectivity for the "Good" wines suggests that the model struggles to differentiate the higher quality wines as well as it does with lower quality ones.

4. K-Nearest Neighbors

We also used the k-nearest neighbors (KNN) algorithm for a classification task, and the process of hyperparameter tuning was conducted using GridSearchCV. GridSearchCV is a systematic approach to hyperparameter optimization that tests various combinations of specified parameters to identify the best-performing configuration. By utilizing cross-validation, it ensures that the performance estimates are reliable and not subject to overfitting, ultimately aiming to maximize the accuracy of the model.

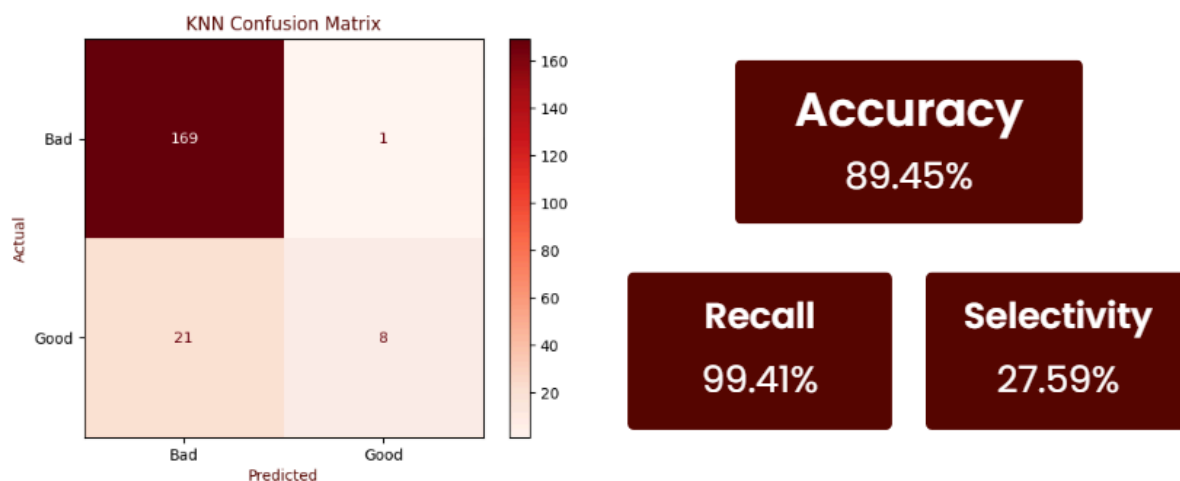
Several key hyperparameters were considered in this tuning process. First, the `n_neighbors` parameter, which determines the number of neighbors considered in the classification, was varied across odd values ranging from 3 to 49 to avoid potential ties. Second, the `weights` parameter, which controls how the contribution of each neighbor is weighted, was explored using two options: 'uniform,' where all neighbors are given equal importance, and 'distance,' where closer neighbors have more influence on the classification. Finally, the `p` parameter was examined to select the appropriate distance metric, with a value of 1 representing the

Manhattan distance and a value of 2 or higher corresponding to Euclidean or other distance metrics.

```
Tuned hyper parameters: {'n_neighbors': 17, 'p': 1, 'weights': 'distance'}  
Best cross-validated accuracy: 0.91
```

After performing the grid search, the optimal hyperparameters were found to be 17 neighbors ($n_neighbors = 17$), the Manhattan distance metric ($p = 1$), and a distance-based weighting scheme ($weights = 'distance'$). This configuration resulted in the highest cross-validated accuracy of 0.91, demonstrating that this set of parameters provided the best classification performance for the task at hand.

Next, we run the k-nearest neighbor with these parameters.



The K-Nearest Neighbor (KNN) model for predicting wine quality shows an overall accuracy of 89.45%, slightly higher than the Random Forest model. The confusion matrix reveals that the model performs exceptionally well in predicting "Bad" quality wines, with 169 out of 170 correctly classified. Only 1 "Bad" wine was misclassified as "Good," highlighting the model's strong ability to identify lower-quality wines.

Despite the selectivity issue, the recall is extremely high at 99.41%, indicating that the KNN model is nearly perfect in identifying "Bad" wines, much like the Random Forest model. However, it struggles to accurately distinguish and identify "Good" wines.